

SUDHEER PALAKURLA

AI Engineer | RAG & LLM Systems | Software Engineer

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Summary:

AI Engineer with 8+ years' full-stack engineering experience and hands-on production work designing and building LLM-powered systems — retrieval-augmented generation (RAG) with vector search, multi-model LLM routing with automatic failover, and agentic AI workflows — end-to-end from architecture through deployment. Strong foundation in Python/FastAPI, Node.js, React/TypeScript and PostgreSQL, with production experience in Model Context Protocol (MCP) integration, prompt engineering, tokenization/context management and LLM evaluation.

Technical & Leadership Expertise:

- **AI & LLM Engineering:** Prompt engineering, RAG architecture (PostgreSQL + pgvector, HNSW indexing), real-time voice AI (OpenAI Realtime API, Gemini Live), multi-model LLM routing & automatic failover, agentic AI workflows, Model Context Protocol (MCP), LLM guardrails & safety evaluation, tokenization & context management, LLM integration & evaluation, AI-assisted development (ChatGPT, Claude & more)
- **Delivery Leadership:** Agile & Scrum, sprint & release planning, estimation, risk and dependency management, roadmap prioritisation, stakeholder reporting, client/vendor coordination
- **Technical Leadership:** Architecture decisions, coding standards, version control, code-review governance, mentoring & onboarding, technical documentation
- **Frontend Engineering:** React.js, Next.js, Angular, TypeScript, JavaScript (ES7+), Redux, React Hooks, Vite, HTML5, CSS3, jQuery, Bootstrap, Material UI, WordPress, WooCommerce, responsive design
- **Backend & Data:** Node.js, Express.js, Python (FastAPI), PHP, REST API design, MySQL, PostgreSQL, MongoDB, SQL, Alembic (schema migrations)
- **Cloud, DevOps & Quality:** AWS, OCI, Cloudflare (Workers, Pages, D1, R2, KV, Workers AI), Docker, Jenkins, GitLab CI/CD, GitHub, Bitbucket, SonarQube, Postman, Jest, React Testing Library, Vitest, Playwright, pytest, BrowserStack, TDD/BDD, WCAG accessibility, performance optimisation
- **Collaboration & Delivery Tools:** Jira, Confluence, Microsoft Teams, Slack, Figma, Adobe Photoshop, Canva

Selected Career Impact:

- Delivered or contributed to 95+ client web and e-commerce builds across UK and international markets.
- Led legacy-to-React/TypeScript/Next.js migrations that lifted application performance and maintainability by 30–40%.
- Architected and shipped RAG-based retrieval systems (PostgreSQL + pgvector, HNSW indexing) across production platforms, powering AI-assisted search and in-app guidance.
- Introduced Vitest/Jest and Playwright test coverage, reducing production defects by ~40%.
- Designed multi-model LLM routing with automatic failover across hosted inference providers, improving reliability of AI-assisted, user-facing features.
- Built a real-time voice AI assistant (OpenAI Realtime API) with automatic fallback to Gemini Live, ensuring uninterrupted voice interactions across provider outages.

Educational Qualification:

- **Master of Computer Science (Data Science & AI) – Bournemouth University, UK | Jan 2020 – Jan 2022**
- **Master of Business Administration (HR & Finance) – Osmania University, India.**
- **Bachelor of Technology (Computer Science Engineering) – Kakatiya University, India.**

Professional Experience:

Web Designer - Brittania Consultancy Services Limited, UK | April 2024 - Current

- Own end-to-end delivery of client web applications — from requirements and technical design through development, testing, deployment and post-launch support — using React.js, TypeScript, Next.js and Vite.
- Act as primary technical point of contact for clients, translating business goals into specifications, wireframes, delivery priorities and acceptance criteria.
- Plan and track delivery in Jira and Confluence, coordinating scope, risks and milestones with clients and stakeholders via Microsoft Teams and Slack.
- Integrate frontend applications with backend REST APIs (Node.js, Python/FastAPI) and enforce code-quality gates using SonarQube alongside Vitest/Jest and Playwright.
- Manage deployment and infrastructure across OCI, AWS, Bluehost and GoDaddy, ensuring high availability across concurrent client projects.
- Designed and built a RAG-based AI assistant for a case-management ERP platform — PostgreSQL + pgvector semantic search with HNSW indexing, multi-model LLM routing across hosted providers and automatic failover to a local model — powering in-app AI search and user guidance.
- Built a serverless client-onboarding platform on Cloudflare Pages and Workers (Hono), using D1 for relational data, KV for low-latency caching and R2 for object storage.
- Integrated Cloudflare Workers AI for in-app AI-assisted features within the same edge architecture, alongside the RAG assistant.
- Apply AI-assisted development practices — prompt engineering and agentic workflows with ChatGPT and Claude — to accelerate delivery while upholding SEO, WCAG accessibility and performance standards.

Independent Project — AI-Driven SaaS Platform (Sole Engineer)

- Owned end-to-end architecture and delivery as sole engineer for an AI-driven SaaS platform — from data model and backend through LLM integration, frontend and deployment.
- Built a RAG-based AI onboarding assistant using PostgreSQL + pgvector with HNSW indexing for semantic retrieval.
- Implemented multi-model LLM routing — selecting across hosted inference providers by language/workload — with automatic failover to a local fallback model for resilience.
- Deployed and operated the platform end-to-end on Oracle Cloud Infrastructure (OCI), managing PostgreSQL schema evolution with Alembic migrations across the application lifecycle.
- Built a real-time voice assistant using the OpenAI Realtime API for live conversational voice interaction, with automatic fallback to Gemini Live to maintain continuity during provider outages.

Senior Frontend Developer - Sharp Gaming Limited, UK | Oct 2022 – June 2023

- Led migration of a legacy JavaScript application to React with TypeScript, improving performance, maintainability and developer productivity.
- Collaborated with product, backend, QA and DevOps teams — coordinated via Jira and Slack — to plan and deliver API-driven features and production fixes.
- Conducted code reviews and mentoring sessions, embedding coding standards and SonarQube quality gates across the frontend team.
- Integrated complex, real-time REST APIs with Redux and React Hooks for state management.

Senior Developer - Crazy Techsol Pvt Ltd, India | May 2017 – Jan 2020

- Led delivery across multiple concurrent client projects, coordinating requirements, frontend development, API integration, testing and stakeholder updates.
- Spearheaded migration of applications from PHP/jQuery to React, TypeScript and Redux, reducing technical debt and improving scalability.
- Built full-stack e-commerce platforms (frontend and backend) for 8+ client stores spanning clothing and bio/wellness products, using PHP for some projects and Node.js for others.
- Established reusable component libraries, code-review practices, version-control standards and CI processes adopted across the team.
- Authored best-practice manuals and technical documentation that streamlined workflows for cross-functional teams.
- Mentored junior developers and led client meetings to align technical delivery with business and commercial objectives.

Web Designer - Crazy Designers, India | Dec 2014 – Apr 2017

- Designed and delivered 40+ responsive websites and e-commerce solutions end-to-end using WordPress, WooCommerce, Magento, HTML5/CSS3, JavaScript and Bootstrap.
- Created UI mockups and visual design assets in Adobe Photoshop, translating them into responsive WordPress and static HTML/CSS website builds.
- Worked directly with clients to gather requirements, build prototypes and translate business needs into practical digital solutions.
- Built payment gateway integrations and custom CMS themes, improving performance, usability and cross-browser compatibility.

Selected Projects

Brittania Consultancy, SK Security Services, Betfred Games, 3 Bolt Court, i20Fever, Dr Foot App, GetMyHotels, Crazy Techsol, KR Energy Consultants, KSRG Energy, Sanguine Bio — full portfolio available on request.